Income, Enforcement Intensity, and Restaurant Inspection Outcomes in NYC

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Policy Questions

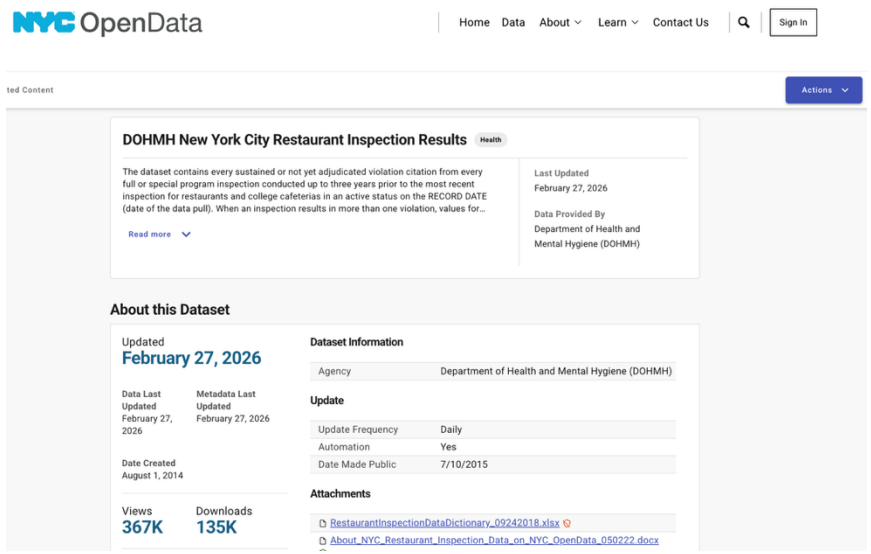
- Are restaurant inspection efforts equitably distributed across neighborhoods by income?
- Is enforcement intensity aligned with observed food-safety outcomes (inspection scores/violations)?
- How do these relationships change over time, and what does that imply for allocating inspection resources?

Data & Units

- DOHMH New York City Restaurant Inspection Results which provides NYC DOHMH restaurant inspection results and violation records.
- ACS 2024 5-year estimates of median household income and total household (by Census tract)
- TIGER/Line Shapefiles (2022): Census tract boundary files for NYC (used for spatial joins and mapping).

Dataset Overview

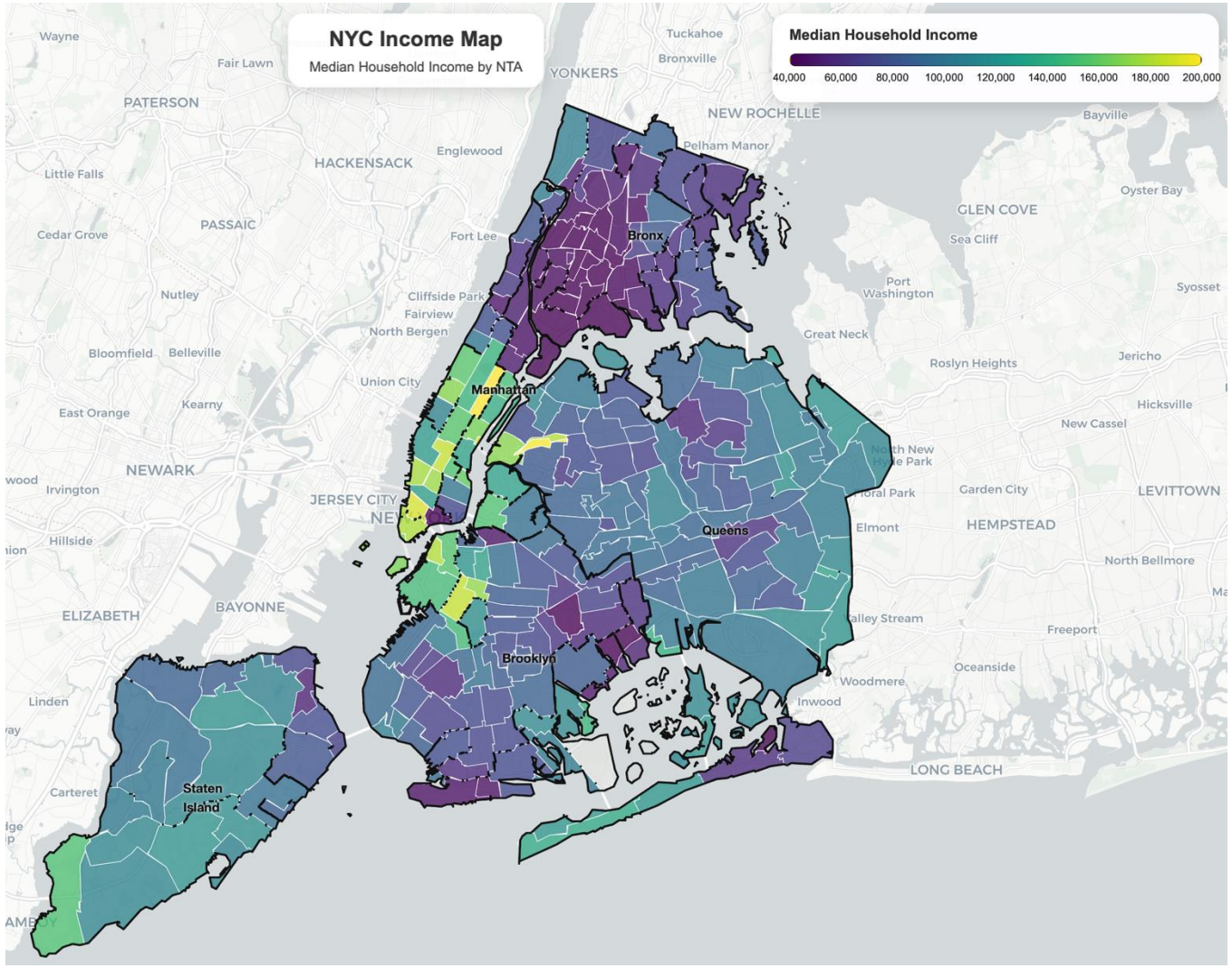
- An inspection is a health department visit that evaluates a restaurant's compliance and produces an overall outcome (score/grade), while a violation (citation) is a specific rule infraction recorded during that inspection—so one inspection can generate multiple violations/citations.
- We mainly use **inspection date** (when it occurred), **inspection type** (what kind of visit), **score** (numeric outcome; higher typically indicates worse conditions), and **location** (where the restaurant is) to analyze how inspection outcomes and enforcement patterns vary across neighborhoods and over time.



Methods

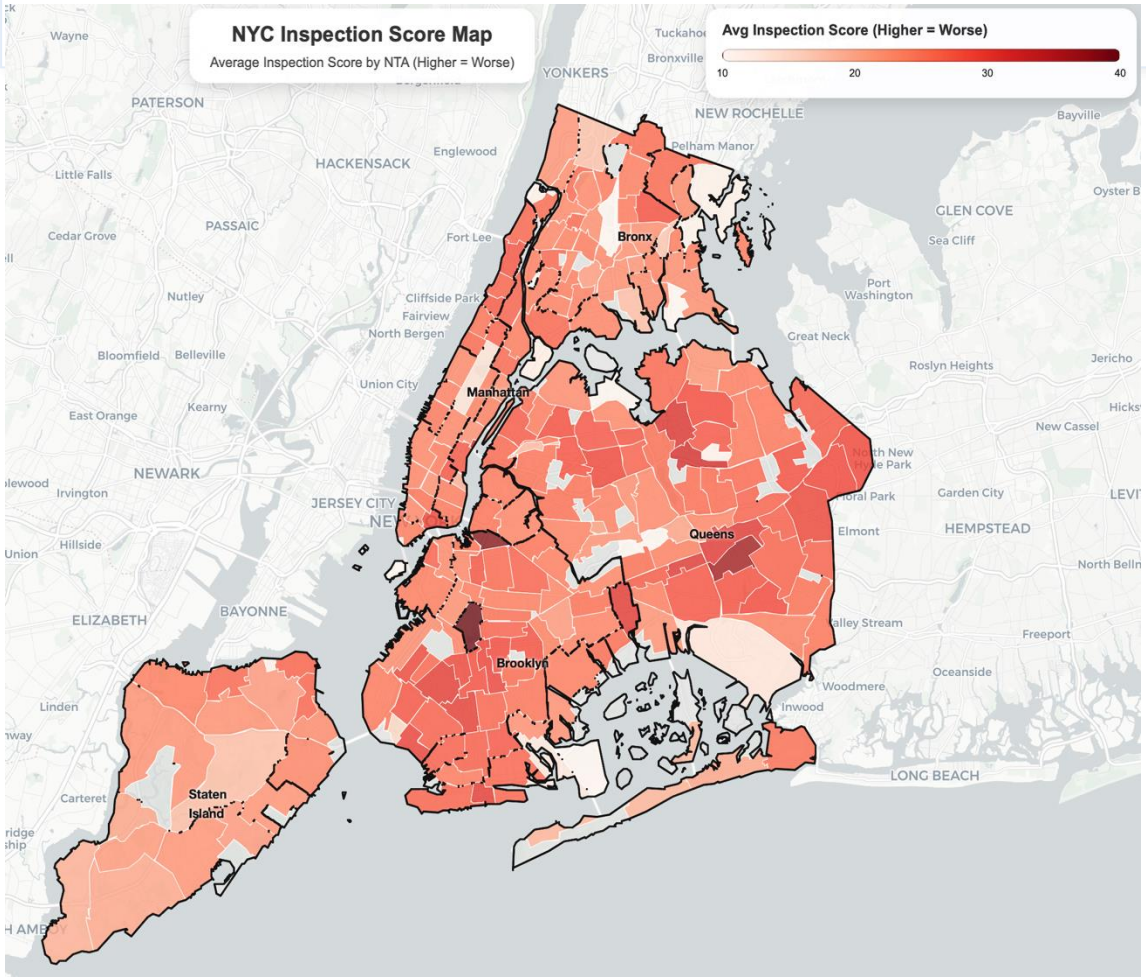
- We integrated NYC restaurant inspection records with neighborhood socioeconomic data by (1) merging tract-level ACS households and median income to census tract polygons, (2) aggregating tract demographics to NTA boundaries via area-weighted overlays, and (3) spatially joining inspection points to NTAs to attach these NTA-level measures to each inspection and to produce NTA-level summary tables for analysis.
- We combine spatial mapping and correlation-based analysis. First, we map key variables (income, enforcement intensity, and inspection outcomes) across NYC NTAs to highlight geographic patterns. Next, we use scatter plots with fitted trend lines to summarize relationships among these measures. Then, we group NTAs into income quintiles to compare enforcement patterns across income levels. Finally, we also examine how the income–enforcement relationship changes over time.

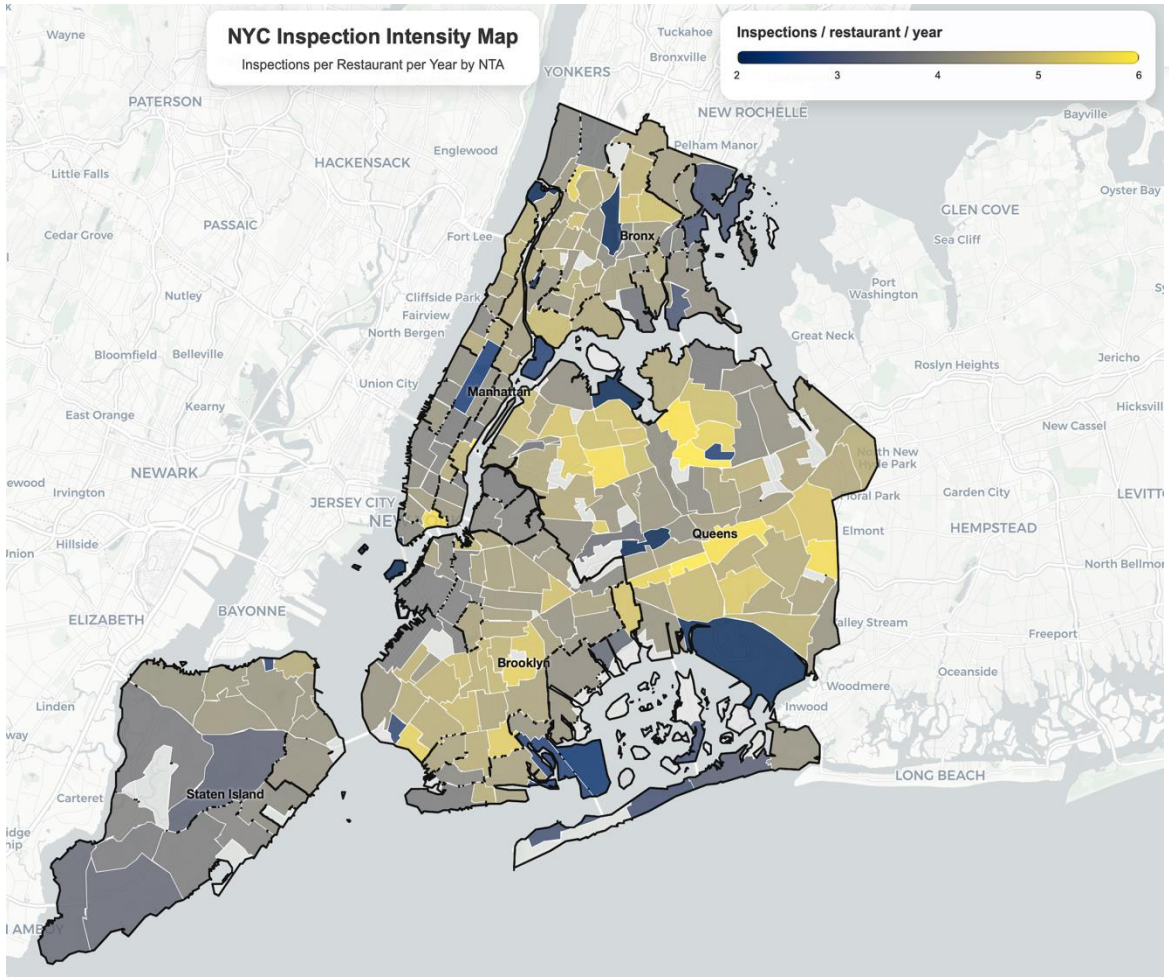
NYC Income Map (Median Household Income by NTA)



- Income is highly spatially clustered across NYC, motivating neighborhood-level comparisons of restaurant inspection outcomes and enforcement intensity.

Outcome & Enforcement Maps

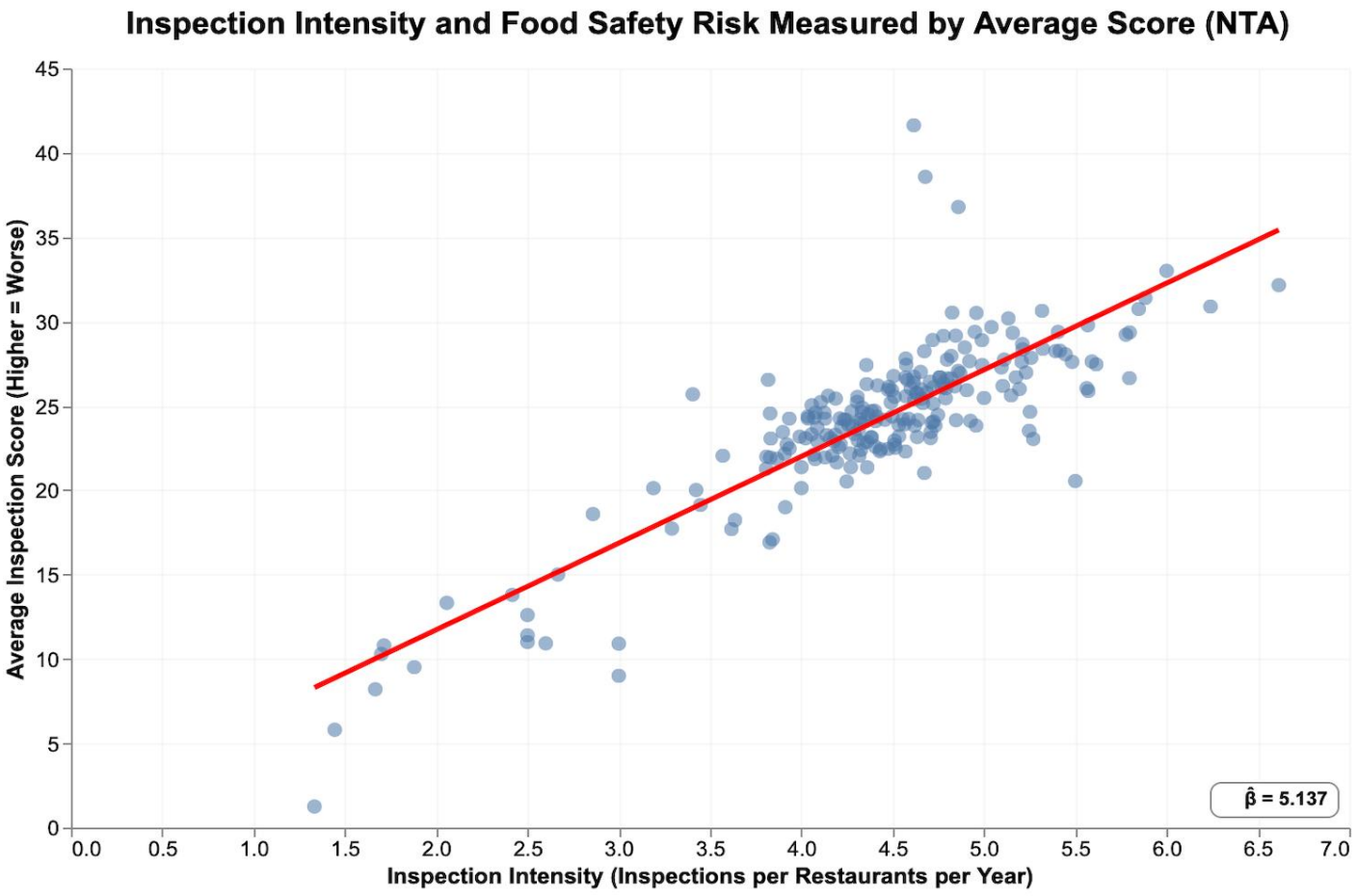




Income vs Enforcement Intensity

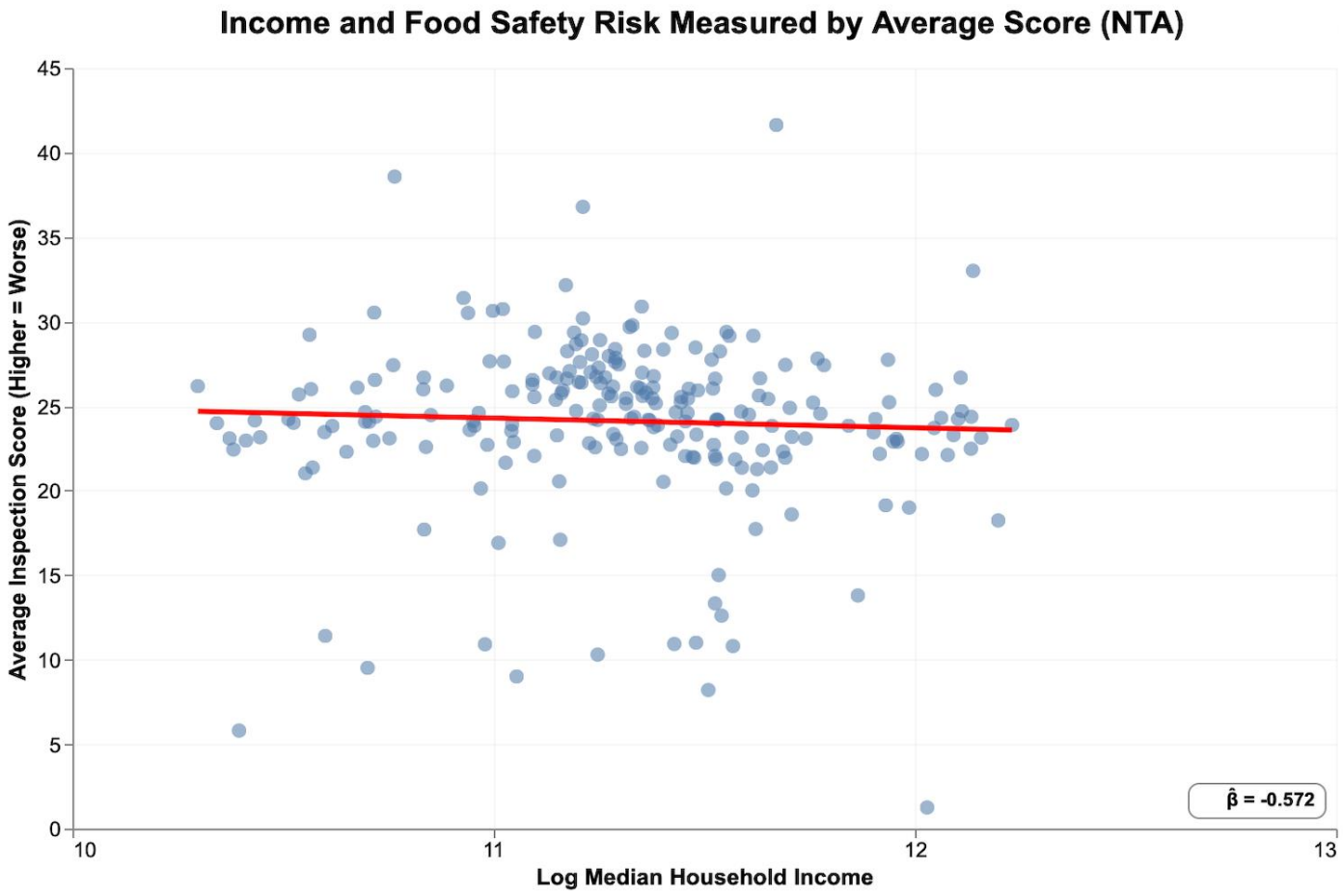
- This scatterplot shows a clear negative association between neighborhood income and inspection intensity: higher-income NTAs tend to receive fewer inspections per restaurant on average.

Enforcement Intensity vs Inspection Score



- This scatterplot shows a positive association between inspection intensity and average inspection score (higher = worse): NTAs with more inspections per restaurant tend to have higher observed scores.

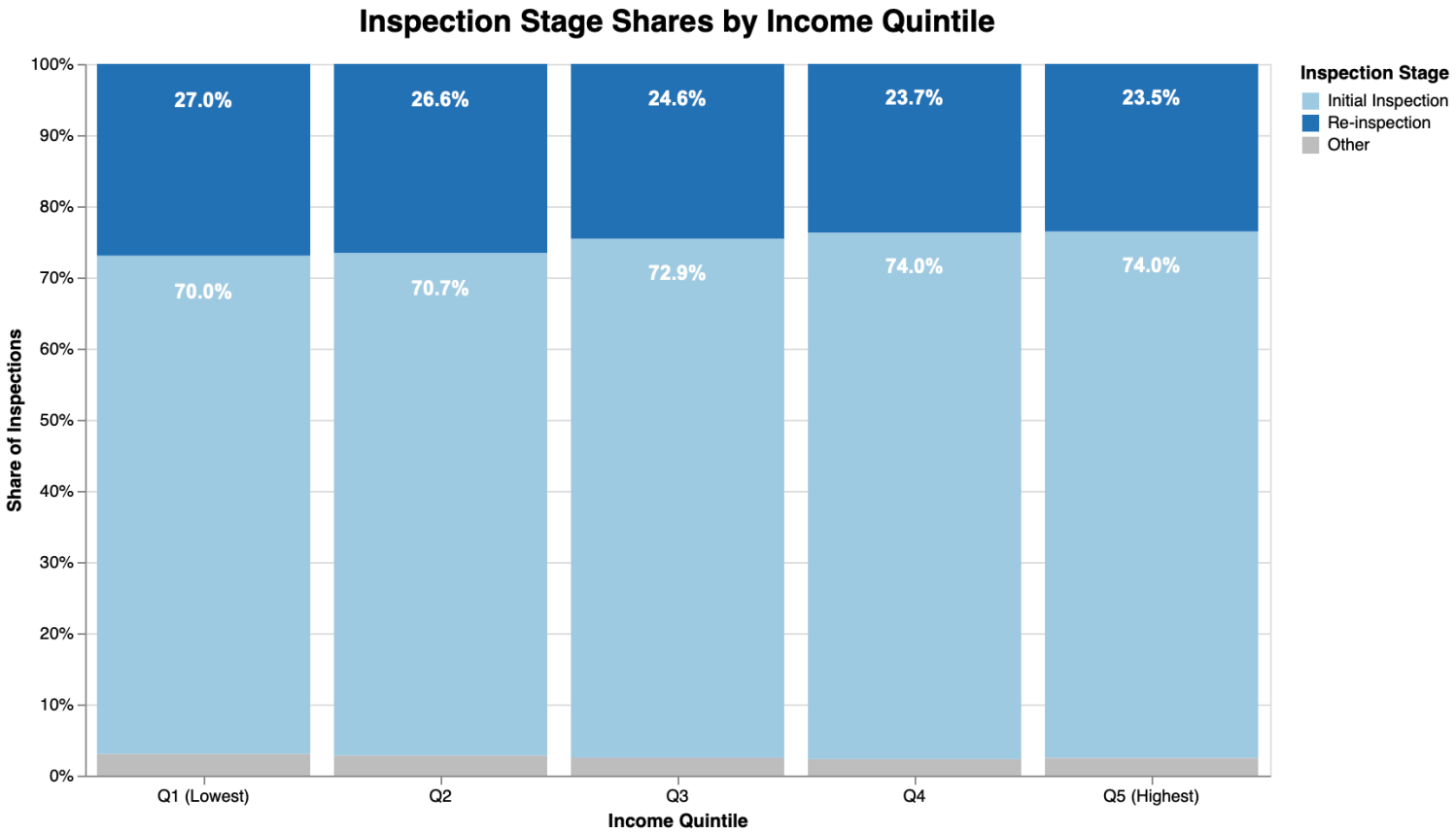
Income vs Inspection Score



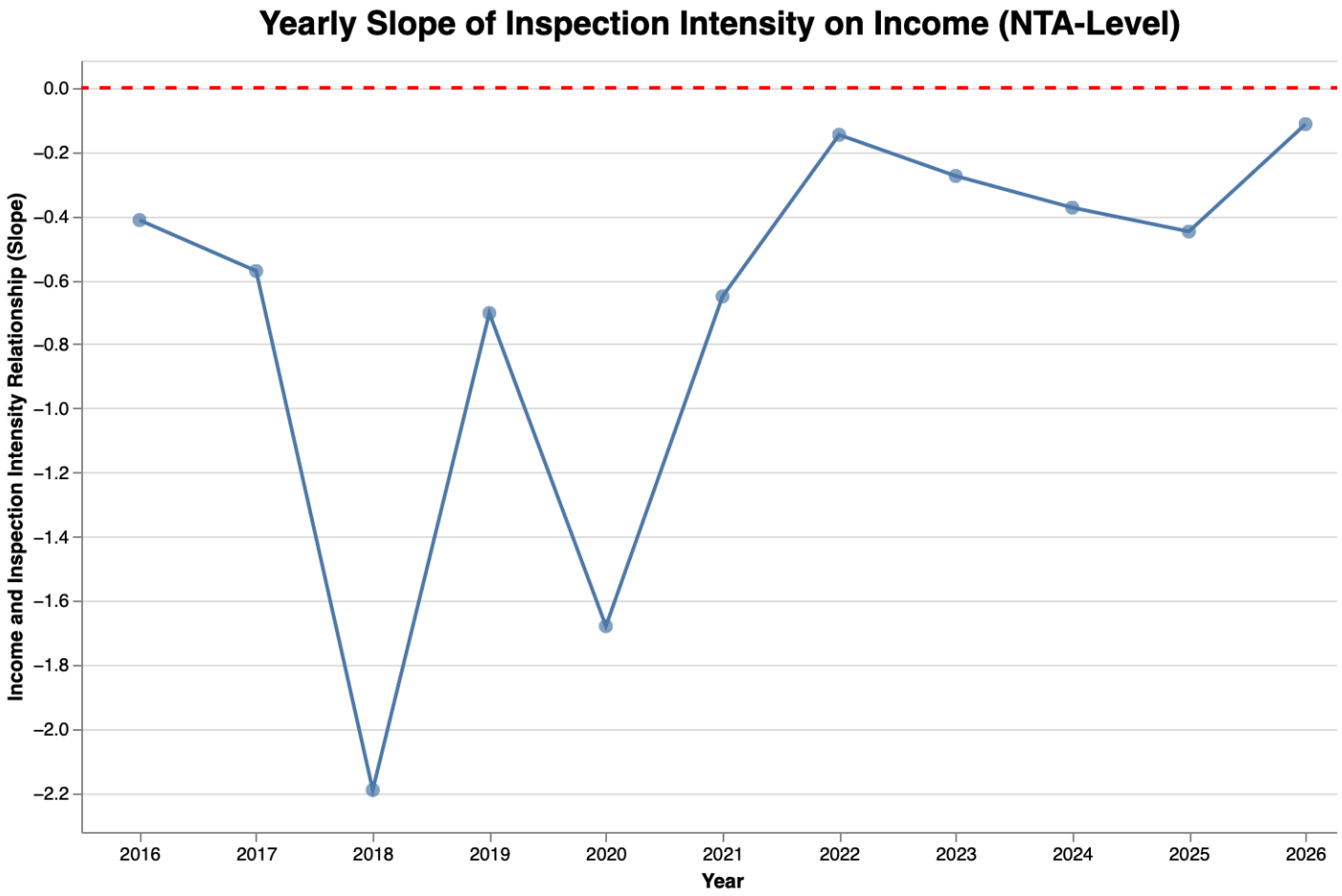
- This plot suggests a weak relationship between neighborhood income and average inspection score: the fitted line is nearly flat, indicating that higher-income NTAs do not consistently have better (lower) scores.

Income Group & Inspection Stage

- Re-inspections are slightly more common in the lowest-income quintile (27% in Q1 vs 23.5% in Q5), though overall inspection-stage shares are broadly similar across income groups.



Relation between Income and Enforcement Intensity Over Time



- This plot shows that the income–enforcement intensity gradient is consistently negative (below the zero line), meaning higher-income NTAs generally have lower inspection intensity.

Streamlit Demo

- https://30538-final-project-h9h7wqmhtuwhwh8nndhnv4.streamlit.app

Policy Implications

- Inspection intensity is not evenly distributed across neighborhood income. This raises concerns about fairness and accountability in how inspection resources are allocated. Agencies should publish clear rules for where inspections and re-inspections are sent and regularly check whether gaps remain after accounting for measured risk.
- Higher inspection intensity is associated with worse observed outcomes. This may reflect targeting higher risk areas or finding more problems when inspections are more frequent. Therefore, outcomes should be reported together with inspection coverage, and enforcement in high need areas should be paired with basic support such as training and outreach.